

Problem

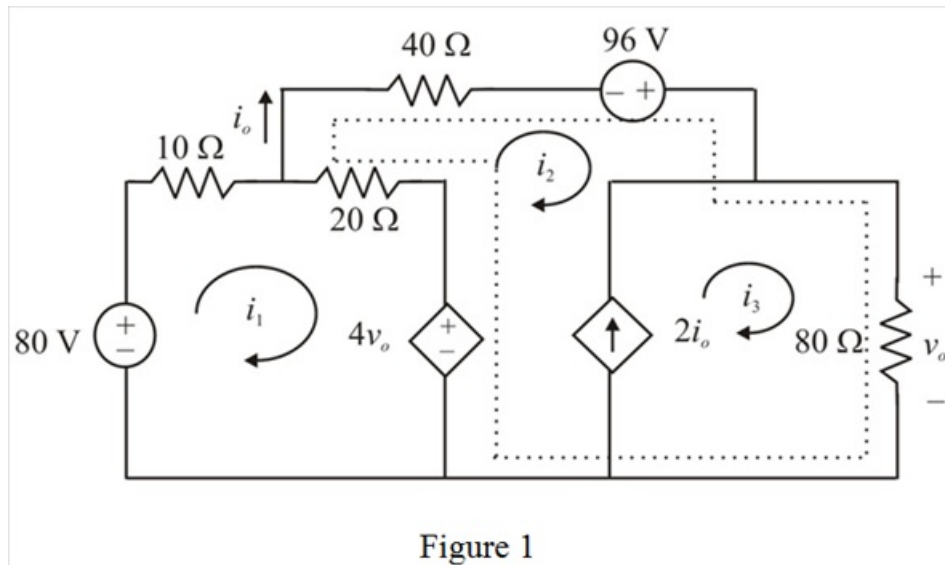
Rework Prob. 3.30 using mesh analysis.

Step-by-step solution

Step 1 of 4

Refer to Figure 3.79 in the textbook.

Draw the circuit diagram with mesh currents.



Step 2 of 4

Apply Kirchhoff's voltage law to Loop 1.

$$-80 + 10i_1 + 20(i_1 - i_2) + 4v_o = 0$$

$$-80 + 10i_1 + 20(i_1 - i_2) + 4(80i_3) = 0$$

$$-80 + 10i_1 + 20i_1 - 20i_2 + 320i_3 = 0$$

$$3i_1 - 2i_2 + 32i_3 = 8 \quad \dots\dots (1)$$

Apply Kirchhoff's voltage law to Loop 2 and Loop 3 together.

$$-4v_o + 20(i_2 - i_1) + 40i_2 - 96 + 80i_3 = 0$$

$$-320i_3 + 20i_2 - 20i_1 + 40i_2 - 96 + 80i_3 = 0$$

$$-20i_1 + 60i_2 - 240i_3 = 96$$

$$-2i_1 + 6i_2 - 24i_3 = 9.6 \quad \dots\dots (2)$$

Step 3 of 4

From the dependent current source,

$$2i_o = i_3 - i_2$$

$$2i_2 - i_3 + i_2 = 0$$

$$3i_2 - i_3 = 0 \dots\dots (3)$$

Solve equations (1), (2) and (3).

$$i_1 = 143.04 \text{ A}$$

$$i_2 = -4.48 \text{ A}$$

$$i_3 = -13.44 \text{ A}$$

Step 4 of 4

Calculate the value of the output current, i_o .

$$i_o = i_2$$

$$= -4.48 \text{ A}$$

Therefore the output current, i_o is -4.48 A.

Calculate the value of the output voltage, v_o .

$$v_o = 80i_3$$

$$= 80(-13.44)$$

$$= -1075.2 \text{ V}$$

Therefore the output voltage, v_o is -1075.2 V.